

SNOWFLAKE COCO CLI HACKATHON - GCC EDITION

SupplyChain Atlas

A governed disruption command center for resilient supply chains

- Snowflake-native
- Synthetic data only
- Explainable decisions

Connect operational signals to evidence-backed recovery action.



Disruptions are visible. Their business impact is not.

Operations teams must reconcile fragmented alerts before they can decide what to escalate.

SYNTHETIC SCENARIO

120 shipments

CRITICAL DISRUPTIONS

10

HIGH-RISK DISRUPTIONS

32

GOVERNED EXPOSURE

USD 541.8M

The operational gap

- **Siloed alerts**

Carrier delay, sensor, supplier, and order data arrive separately.

- **Opaque prioritization**

Teams cannot reliably connect an alert to customer exposure.

- **Weak audit trail**

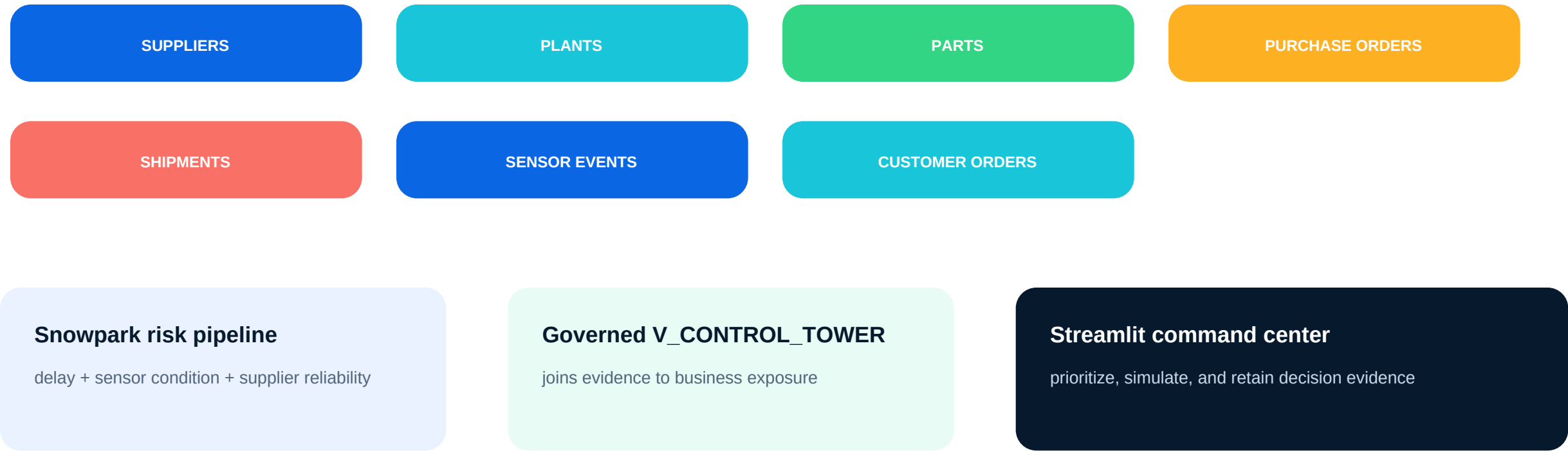
Stakeholders need the factors and source entities behind a decision.

Atlas turns the question into an action

Which shipment threatens customer commitments this week, why is it risky, and what should we do first?

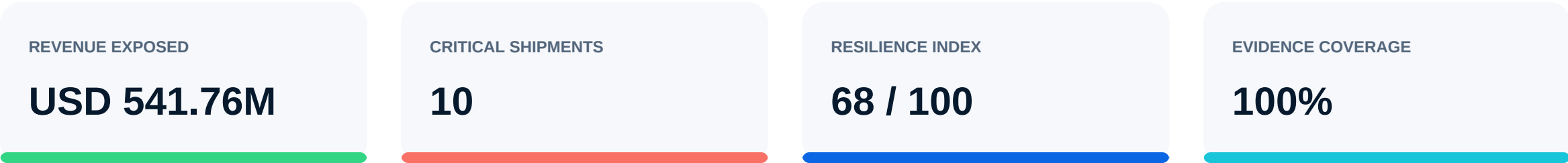
One governed ontology, one decision surface

Every disruption is represented as a connected operational story.



From input to transparent prioritization

The live Snowflake Streamlit app keeps decision context visible.



Capability 1 - Command center

Ranks critical disruptions by risk score and revenue impact. Supplier concentration exposes where recovery work should begin.

Operator view

- Shipment and risk score
- Part, plant, and supplier
- Revenue at risk and ETA
- Evidence-ready source trail

A dashboard that supports a defensible intervention

Three capabilities turn disruption data into a governed action.

What-if Lab

Simulate recovered days for SHP-0003 and see projected risk reduction.

CAPABILITY 02

Decision Receipt

Name the governed sources and make the risk formula auditable.

CAPABILITY 03

CoCo-assisted workflow

Use CoCo to inspect the governed contract and plan improvements around real Snowflake objects.

CAPABILITY 04

Input: governed operational evidence -> Processing: transparent risk -> Output: ranked recovery action

Built to be reviewed, reproduced, and extended

The project is live, documented, and backed by a public GitHub repository.

Governance by design

- **Synthetic-only scenario**

No customer or proprietary records are used.

- **Inspectable model**

Risk is traceable to delay, sensor, and supplier factors.

- **Evidence contract**

V_CONTROL_TOWER connects recommendations to named source entities.

Submission assets

- **Live app**

SUPPLYCHAIN_ATLAS.CORE.SUPPLYCHAIN_ATLAS

- **Repository**

github.com/manoharpeddinti-embedded-engineer/supplychain-atlas

- **Reproduction**

SQL setup, generator, Snowpark pipeline, semantic model, Streamlit app

Detect the disruption. Quantify the impact. Rehearse the recovery. Prove the decision.